10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 1 — Complex Numbers

Unit 1 | MCQs + Review Exercise | New Book (PECTAA)

Review Exercise 1 (Solved)

Q.1 Choose the correct option.

(i) $i^2 + i^4 =$

(a) -1 (b) 0 (c) 1 (d) 2

Answer: (b) 0 $i^2 = -1$ and $i^4 = 1$, so $i^2 + i^4 = -1 + 1 = 0$.

(ii) Real part of $(2 - 3i)(2 + 3i)$ is:

(a) -3 (b) 1 (c) 4 (d) 13

Answer: (d) 13 $(2 - 3i)(2 + 3i) = 4 - 9i^2 = 4 + 9 = 13$, so real part = 13.

(iii) Imaginary part of $(2 - i)(2 + i)$ is:

(a) 0 (b) 1 (c) 7 (d) 9

Answer: (a) 0 $(2 - i)(2 + i) = 4 - i^2 = 5 = 5 + 0i$, so imaginary part = 0.

(iv) $x + iy$ will be a pure imaginary number, when:

(a) $y = 0$ (b) $x = 0$ (c) $i = 0$ (d) $x = 0, y = 0$

Answer: (b) x = 0 A pure imaginary number has zero real part, so $x = 0$ (and $y \neq 0$).

(v) What is the additive inverse of $5 - 2i$?

(a) $5 + 2i$ (b) $-5 - 2i$ (c) $5 - 2i$ (d) $-5 + 2i$

Answer: (d) −5 + 2i $-(5 - 2i) = -5 + 2i$, and $(5 - 2i) + (-5 + 2i) = 0$.

(vi) What is the multiplicative inverse of $z = 1 + i$?

(a) $1 - i$ (b) i (c) $1/2 - 1/2\,i$ (d) $-1/\sqrt{2} + 1/\sqrt{2}\,i$

Answer: (c) 1/2 − 1/2 i $z^{-1} = 1/(1 + i) = (1 - i)/2 = 1/2 - 1/2\,i$.

(vii) If $z = 4 - 3i$, then $z\,\bar{z} =$

(a) 3 (b) 9 (c) 16 (d) 25

Answer: (d) 25 $z\,\bar{z} = (4)^2 + (3)^2 = 16 + 9 = 25$.

(viii) Conjugate of $9 - 4i$ is:

(a) $-9 - 4i$ (b) $9 + 4i$ (c) $9 + 9i$ (d) $4 - 9i$

Answer: (b) 9 + 4i In the conjugate the sign of the imaginary part is changed.

(ix) If $z = 4 + 4i$, then $z + \bar{z} =$

(a) 8 (b) $8 + 8i$ (c) $8i$ (d) 0

Answer: (a) 8 $\bar{z} = 4 - 4i$, so $z + \bar{z} = (4 + 4i) + (4 - 4i) = 8$, a real number.

(x) If $z = 5 + 4i$, then $|z| =$

(a) 9 (b) 25 (c) 41 (d) $\sqrt{41}$

Answer: (d) √41 $|z| = \sqrt{(5)^2 + (4)^2} = \sqrt{25 + 16} = \sqrt{41}$.

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Q.2 Short questions

(i) Is "0" a complex number? Explain.

Yes, 0 is a complex number, because every complex number is written as a + bi. We can write 0 = 0 + 0i, in which the real part is 0 and the imaginary part is 0. For this reason 0 is called a complex number as well as a real number.

(ii) What is the result of multiplying a complex number by its conjugate?

The result is always a real number, and it is equal to the square of the modulus of that complex number, that is z · z̄ = |z|^{2}. For example, (2 + 3i)(2 − 3i) = 4 + 9 = 13, which is a real number.

(iii) State the condition for two complex numbers to be equal.

Two complex numbers a + bi and c + di are equal only when their real parts are equal and their imaginary parts are equal, that is a = c and b = d.

Q.3 Simplify

(i) i^{37}

(ii) $i^{13} \times i^{11}$

(iii) $(-i)^{-9}$

(iv) $(3 - 4i)(5 - 6i)$

(v) $(3 + 4i) + (5 - 7i)$

(i) i^{37}

$i^{37} = i^{36} \times i$

$= (i^4)^9 \times i$

$= (1)^9 \times i$

$= i$

Answer: i

(ii) $i^{13} \times i^{11}$

$i^{13} \times i^{11} = i^{13+11}$

$= i^{24}$

$= (i^4)^6$

$= (1)^6$

$= 1$

Answer: 1

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(iii) $(-i)^{-9}$

$(-i)^9 = (-1)^9 \times i^9$

$= -(i^8 \times i)$

$= -(1 \times i)$

$= -i$

$\therefore (-i)^{-9} = \frac{1}{-i}$

$= \frac{1}{-i} \times \frac{i}{i}$

$= \frac{i}{-i^2}$

$= \frac{i}{1}$

$= i$

Answer: i

(iv) (3 − 4i)(5 − 6i)

$= 15 - 18i - 20i + 24i^2$

$= 15 - 38i + 24(-1)$

$= 15 - 38i - 24$

Answer: −9 − 38i

(v) (3 + 4i) + (5 − 7i)

$= (3 + 5) + (4 - 7)i$

Answer: 8 − 3i

Q.4 Find additive and multiplicative inverse of z = 8 − 9i

(i) Additive inverse

$-z = -(8 - 9i)$

$= -8 + 9i$

Answer: −8 + 9i

(ii) Multiplicative inverse

$z^{-1} = \frac{1}{8 - 9i} \times \frac{8 + 9i}{8 + 9i}$

$= \frac{8 + 9i}{(8)^2 + (9)^2}$

$= \frac{8 + 9i}{64 + 81} = \frac{8 + 9i}{145}$

$= \frac{8}{145} + \frac{9}{145} i$

Answer: 8/145 + 9/145 i

Q.5 If $z_1 = 3 + 4i$ and $z_2 = 2 + 3i$, then verify that:

(i) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

(ii) $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$

(iii) $\overline{z_1/z_2} = \overline{z_1} / \overline{z_2}$

(iv) $|z_1| = |-z_1|$

(v) conjugate of conjugate of $z_2 = z_2$

(vi) $z_1 \cdot \overline{z_1} = |z_1|^2$

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(i) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

$z_1 + z_2 = (3 + 4i) + (2 + 3i)$

$= 5 + 7i$

L.H.S. $= 5 - 7i$

R.H.S. $= (3 - 4i) + (2 - 3i)$

$= 5 - 7i$

Answer: L.H.S. = R.H.S., hence verified

(ii) $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$

$z_1 z_2 = (3 + 4i)(2 + 3i)$

$= 6 + 9i + 8i + 12i^2$

$= -6 + 17i$

L.H.S. $= -6 - 17i$

R.H.S. $= (3 - 4i)(2 - 3i)$

$= 6 - 9i - 8i + 12i^2$

$= -6 - 17i$

Answer: L.H.S. = R.H.S., hence verified

(iii) $\overline{z_1/z_2} = \overline{z_1} / \overline{z_2}$

$\frac{z_1}{z_2} = \frac{3 + 4i}{2 + 3i} \times \frac{2 - 3i}{2 - 3i}$

$= \frac{6 - 9i + 8i - 12i^2}{4 + 9}$

$= \frac{18 - i}{13}$

L.H.S. $= \frac{18 + i}{13}$

R.H.S. $= \frac{3 - 4i}{2 - 3i} \times \frac{2 + 3i}{2 + 3i}$

$= \frac{6 + 9i - 8i - 12i^2}{4 + 9}$

$= \frac{18 + i}{13}$

Answer: L.H.S. = R.H.S., hence verified

(iv) $|z_1| = |-z_1|$

$|z_1| = \sqrt{(3)^2 + (4)^2}$

$= \sqrt{9 + 16}$

$= \sqrt{25}$

$= 5$

$-z_1 = -3 - 4i$

$|-z_1| = \sqrt{(-3)^2 + (-4)^2}$

$= \sqrt{25}$

$= 5$

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Answer: $|z_1| = |-z_1|$, hence verified

(v) Conjugate of the conjugate equals the number itself

$\overline{z_2} = 2 - 3i$

Taking the conjugate again: $\overline{\overline{2 - 3i}} = 2 + 3i$

$= z_2$

Answer: conjugate of the conjugate = z_2, hence verified

(vi) $z_1 \cdot \overline{z_1} = |z_1|^2$

$z_1 \cdot \overline{z_1} = (3 + 4i)(3 - 4i)$

$= (3)^2 - (4i)^2$

$= 9 + 16$

$= 25$

$|z_1|^2 = (\sqrt{25})^2$

$= 25$

Answer: $z_1 \cdot \bar{z}_1 = |z_1|^2$, hence verified

Q.6 If $z_1 = 5 + 4i$, $z_2 = 3 + 2i$, then find:

(i) z_1z_2 (ii) z_1/z_2

(iii) $\overline{z_1} \cdot \overline{z_2}$ (iv) $|z_1z_2|$

(i) z_1z_2

$z_1z_2 = (5 + 4i)(3 + 2i)$

$= 15 + 10i + 12i + 8i^2$

$= 15 + 22i - 8$

Answer: 7 + 22i

(ii) z_1/z_2

$$\frac{z_1}{z_2} = \frac{5 + 4i}{3 + 2i} \times \frac{3 - 2i}{3 - 2i}$$

$$= \frac{15 - 10i + 12i - 8i^2}{9 + 4}$$

$$= \frac{15 + 2i + 8}{13}$$

$$= \frac{23 + 2i}{13}$$

Answer: 23/13 + 2/13 i

(iii) conjugate of z_1 times conjugate of z_2

$\overline{z_1} \cdot \overline{z_2} = (5 - 4i)(3 - 2i)$

$= 15 - 10i - 12i + 8i^2$

$= 15 - 22i - 8$

Answer: 7 − 22i

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(iv) $|z_1z_2|$

$z_1z_2 = 7 + 22i$

$|z_1z_2| = \sqrt{(7)^2 + (22)^2}$

$= \sqrt{49 + 484}$

Answer: $\sqrt{533}$

Q.7 Find real and imaginary parts of $z = (2 + 7i)^{-1}$

$z = \frac{1}{2 + 7i} \times \frac{2 - 7i}{2 - 7i}$

$= \frac{2 - 7i}{(2)^2 + (7)^2}$

$= \frac{2 - 7i}{4 + 49} = \frac{2 - 7i}{53}$

$= \frac{2}{53} - \frac{7}{53} i$

Answer: Re = 2/53, Im = −7/53

Q.8 Solve for z and w:

$iz + (2 - i)w = 4 + i$ ... (1)

$iz + (3 + i)w = 3 + 3i$ ... (2)

Subtract (1) from (2):

$[(3 + i) - (2 - i)]w = (3 + 3i) - (4 + i)$

$(1 + 2i)w = -1 + 2i$

$w = \frac{-1 + 2i}{1 + 2i} \times \frac{1 - 2i}{1 - 2i}$

$= \frac{-1 + 2i + 2i - 4i^2}{1 + 4}$

$= \frac{3 + 4i}{5}$

$w = \frac{3}{5} + \frac{4}{5} i$

Put in (1): $iz = 4 + i - (2 - i)w$

$(2 - i)(\frac{3 + 4i}{5}) = \frac{6 + 8i - 3i - 4i^2}{5}$

$= \frac{10 + 5i}{5}$

$= 2 + i$

$iz = 4 + i - (2 + i)$

$= 2$

$z = \frac{2}{i} \times \frac{i}{i}$

$= \frac{2i}{-1}$

$= -2i$

Answer: z = −2i, w = 3/5 + 4/5 i

Q.9 Solve $(3 - 4i)(a + bi) = 1 + 0i$ and find a and b.

$a + bi = \frac{1}{3 - 4i} \times \frac{3 + 4i}{3 + 4i}$

$= \frac{3 + 4i}{(3)^2 + (4)^2}$

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$= \frac{3 + 4i}{9 + 16} = \frac{3 + 4i}{25}$

$= \frac{3}{25} + \frac{4}{25} i$

Comparing: $a = \frac{3}{25}$, $b = \frac{4}{25}$

Answer: a = 3/25, b = 4/25

Q.10 Solve the equation for x and y:

$(3 - 2i)(x + yi) = 2(x - 2yi) + 2i - 1$

L.H.S. $= 3x + 3yi - 2xi - 2yi^2$

$= (3x + 2y) + (3y - 2x)i$

R.H.S. $= 2x - 4yi + 2i - 1$

$= (2x - 1) + (2 - 4y)i$

Real parts: $3x + 2y = 2x - 1$

$x + 2y = -1$ ... (1)

Imaginary parts: $3y - 2x = 2 - 4y$

$-2x + 7y = 2$ ... (2)

From (1): $x = -1 - 2y$

Put in (2): $-2(-1 - 2y) + 7y = 2$

$2 + 4y + 7y = 2$

$11y = 0$

$y = 0$

$x = -1 - 2(0)$

$= -1$

Answer: x = −1, y = 0

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